

Violin Monica

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SUMMARY

Analytical second-year Computer Science student at Universitas Indonesia with demonstrated technical expertise and leadership experience. Hands-on experience in full-stack development, robotics programming, and cybersecurity, complemented by strong communication skills evidenced by debate competition success.

EXPERIENCE

Teaching Assistant of Programming Foundations 1

Feb 2026 – Jun 2026

- Supervising programming lab sessions
- Developed problems and sample solutions for programming lab exercises
- Graded and evaluated student quizzes and programming lab results

Business and Partnerships Bureau, Student Executive Board of the Faculty of Computer Science, Universitas Indonesia

Sept 2025 – Nov 2025

Software Engineer Intern

- Gained practical experience in front-end development with Next.js, backend API testing with Postman, and version control using GitHub

EDUCATION

Universitas Indonesia, Faculty of Computer Science

First-year Student in Computer Science | Depok, IDN

August 2025 – present

- Academic interests: Cyber Security, Data Science, Software Engineering, and Robotics

SMAS Kristen Bina Kasih Jambi

Student in Engineering Course | Jambi, IDN

July 2022 – May 2025

- Relevant coursework: Advanced Math, Informatics

ORGANIZATIONAL ACTIVITIES & COMMITTEES

University of Indonesia Robotic Team

March 2026 - present

Member of ReignBlaze (Indonesian SAR Robot Contest Team) Programming Division

- Contributed to developing the control program for a line-following robot using Arduino microcontroller in a friendly inter-team competition.
- Contributing to the development of a V-SLAM localization system, integrating an Intel RealSense T265 camera for autonomous robot mapping.
- Contributing to the development of 3D visualizations using RViz to monitor real-time telemetry and sensor fusion data.

RISTEK

Member of Network, Security, and Operating System

- Represented University of Indonesia in national-level cybersecurity competitions to strengthen the organization's technical portfolio.
- Developed Capture The Flag (CTF) challenges for the RISTEK Bobol website, specializing in the Jail and Reverse Engineering categories.

PROJECTS

SORTIFY – AI WASTE IDENTIFICATION WEB APPLICATION

May 2026

Backend Developer

- Architected the backend infrastructure using Python and Django, integrating a CNN (Convolutional Neural Network) for automated waste classification.

- Developed RESTful APIs to bridge deep learning models with a React/Vite frontend, facilitating real-time disposal tips and location-based mapping.
- Implemented geolocation services to connect users with the nearest recycling and disposal facilities.

SAFESTEP – AI-BASED STREET RISK SCORING WEB APPLICATION

Jul 2026

Machine Learning Engineer

- Conducted end-to-end data analysis, feature engineering, model development, and evaluation for an AI-based risk scoring system for women's street safety.
- Designed and implemented RESTful APIs to serve machine learning models for application integration.

NUSA-CROP – AI-BASED PLANT RECOMMENDATION SYSTEM FOR MARGINAL FIELDS WEB APPLICATION

Jul 2026

- Developing an AI-powered crop recommendation system by conducting data preparation, machine learning model development, and RESTful API implementation.
- Contributed to stakeholder requirements analysis, application workflow design, and UI development, including the landing page.

ACHIEVEMENT AND AWARD

FINALIST OF THE ACE CAPTURE THE FLAG

National

Issued by Departemen Teknik Komputer Universitas Diponegoro

August 2026

- Part of a three-member team that advanced to the final round of Capture The Flag, specializing in web exploitation and reverse engineering category.

RUNNER UP OF INFORMATICS NATIONAL SCIENCE OLYMPIAD 2024

Province

Issued by Dinas Pendidikan Provinsi Jambi

March 2024 – May 2024

- Advanced to provincial level: competed against 110 high school students in city level
- Secured 2nd place in provincial level

RUNNER UP OF NATIONAL SCHOOLS DEBATING CHAMPIONSHIP 2024

City

Issued by Dinas Pendidikan Provinsi Jambi

May 2024

- Part of a three-member team that earned second place in the 2024 National Schools Debating Championship city level, surpassing more than five competing teams.

ADDITIONAL

- **Skills:** C++, Python, Canva, Editing, Public Speaking, Arduino IDE, Problem Solving, Communication, Writing, Figma, Next.js, Django, Java, Linux
- **Languages:** Fluent level or native level in Indonesia; Working level proficiency in English; Basic proficiency in Mandarin